WORK AND ENERGY

The differential equation of rectilinear motion is $\frac{W}{g}\ddot{x} = \Sigma F_x$

$$\frac{W}{g}v\frac{dv}{dx} = \Sigma F_x$$

$$\frac{W}{g}\;v\,dv = \Sigma F_x\,dx$$

$$\frac{W}{g}d\left(\frac{v^2}{2}\right) = \Sigma F_x\,dx$$

$$d\left(\frac{W}{g}\frac{v^2}{2}\right) = \Sigma F_x\,dx$$

The term in the bracket $\frac{W}{g}\frac{v^2}{2}$ is known as the kinetic energy of the body. The above equation states that the differential change in the kinetic energy of the particle is equal to the work done by the external force ΣF_x over a small displacement dx

Integrating both sides knowing that the body starts from origin with velocity $\dot{x}_0$

$$\frac{W}{g}\int_{\dot{x}_0}^{\dot{x}} d\left(\frac{v^2}{2}\right) = \int_0^x \Sigma F_x\,.dx$$

$$\frac{W}{g}\frac{\dot{x}^2}{2} - \frac{W}{g}\frac{\dot{x}_0{}^2}{2} = \int_0^x \Sigma F_x\,.dx$$

This equation shows that the change in kinetic energy of the particles is equal to the net work done by ΣF_x over the displacement $0 \to x$. This is known as work-energy principle.

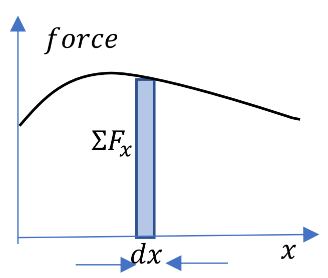
We can also write

$$\frac{W}{g}\frac{\dot{x}_0{}^2}{2} + \int_0^x \Sigma F_x\,.dx = \frac{W}{g}\frac{\dot{x}^2}{2}$$

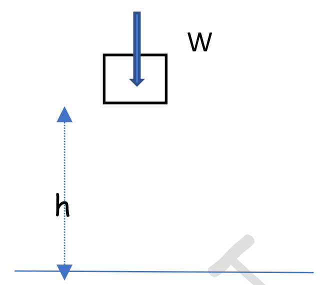
This means that

Initial kinetic energy + Net work done= Final kinetic energy.

We can evaluate $\int_0^x \Sigma F_x\,.dx$ only if ΣF_x is a constant or a function of x. If ΣF_x is a function of x, we can represent it by a force-displacement diagram. Let ΣF_x be the value of the force at a displacement x. consider an infinitesimal displacement dx. Then $\Sigma F_x.dx$ represents the area of the infinitesimal element. Therefore, $\int_0^x \Sigma F_x\,.dx$ represents the area under the displacement diagram. The work done can be evaluated by the area of the force-displacement diagram.



Consider a body of weight W falling freely through a height h. Neglecting air resistance, the only force acting in the direction of motion is weight W.
The initial kinetic energy of the body=0.
Let $\dot{x}$ be the velocity that the body acquires after falling through a height h.
The final kinetic energy of the body $= \frac{W}{g}\frac{\dot{x}^2}{2}$



The change in kinetic energy $= \frac{W}{g}\frac{\dot{x}^2}{2} - 0 = \frac{W}{g}\frac{\dot{x}^2}{2}$

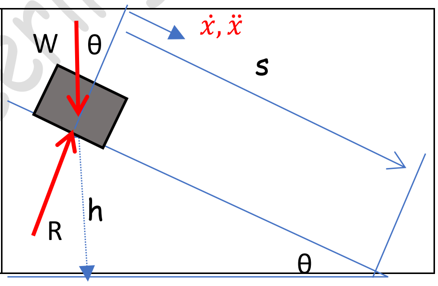
Net work done by weight $= Wh$

By work energy principle, change in kinetic energy= net work done.

$$\frac{W}{g}\frac{\dot{x}^2}{2} = Wh$$

$$\dot{x} = \sqrt{2gh}$$

Consider a body of weight W sliding without friction through a distance s. Neglecting air resistance, the only force acting in the direction of motion is the component of weight W. The initial kinetic energy of the body=0. Let $\dot{x}$ be the velocity that the body acquires after sliding through a distance s.
The final kinetic energy of the body $= \frac{W}{g}\frac{\dot{x}^2}{2}$



The change in kinetic energy $= \frac{W}{g}\frac{\dot{x}^2}{2} - 0 = \frac{W}{g}\frac{\dot{x}^2}{2}$

Net work done by the forces in the direction of motion= $W\sin\theta \,.s$

By work energy principle, change in kinetic energy= net work done.

$$\frac{W}{g}\frac{\dot{x}^2}{2} = W\, s.\sin\theta = W.h$$

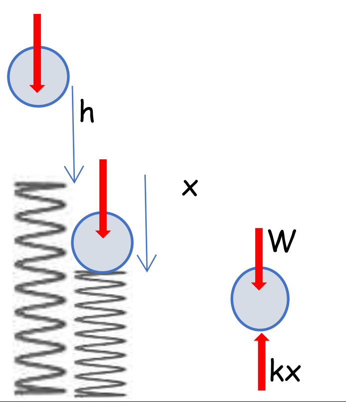
$$\dot{x} = \sqrt{2gh}$$

That is, the velocity acquired by the body after sliding through a friction less inclined plane is same as the velocity it would have acquired in falling freely through a height h

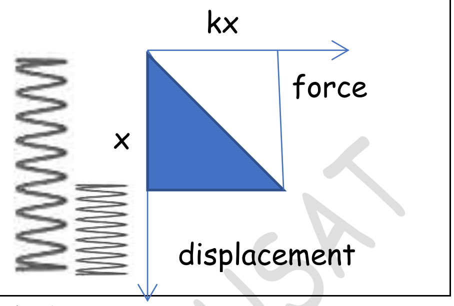
Example 1

A ball of weight $W = 50\ N$ rests on spring of constant k, it produces a static deflection of $2.5\ cm$. How much will the same ball compress the spring if it is dropped from a height of $30\ cm$? Neglect the mass of the spring.

Weight $W = 50\ N$ produces a static deflection of $2.5\ cm$ of the spring.
Therefore, the spring constant $k = \frac{50\ N}{2.5\times10^{-2}\ m} = 2000\ N/m$
The ball is released from rest.
It falls through a height h, then compresses the spring to a maximum distance of x and comes to rest for a moment and then rebounds. Let us consider the motion of the ball from the moment it is released till it compresses the spring to the maximum.



Initial kinetic energy of the ball =0
Final kinetic energy of the ball =0
Change in kinetic energy =0
Only two forces are acting on the system, the weight and the spring force
Work done by the weight= $W(h+x) = 50(0.3+x)$
Work done by the spring force= $-½\, kx^2$ (-ve because force is opposite to the displacement)
Net work done $= 50(0.3+x) - 1000x^2$



By Work- Energy principle, the change in kinetic energy= net work done

$$0 = 50(0.3+x) - 1000x^2$$
$$1000x^2 - 50x - 15 = 0$$
$$x = \frac{50 \pm \sqrt{50^2 + 4 \times 1000 \times 15}}{2000} = \frac{50 \pm 250}{2000} = 0.15\, m\ (+ve\ value)$$

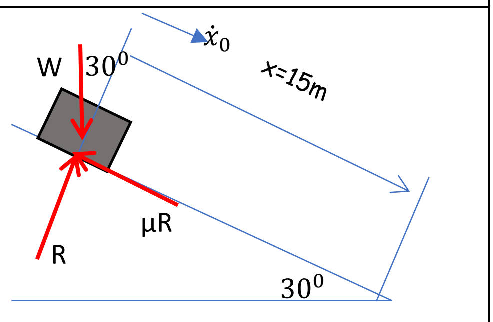
Example 2

A small block of weight $W = 45\ N$ is given an initial velocity $\dot{x}_0 = 3\ m/s$ down a 30^0 inclined plane. If the coefficient of friction between the plane and the block is 0.3, determine its velocity after it had travelled a distance $x = 15m$

The initial velocity of the block down the plane,
$$\dot{x}_0 = 3\ m/s$$
The initial kinetic energy= $\frac{W}{g}\frac{\dot{x}_0{}^2}{2} = \frac{45}{9.81} \times \frac{3^2}{2} = 20.64\ J$

Let $\dot{x}$ be the velocity of the block after it has travelled a distance of 15 m.

Final kinetic energy of the block= $\frac{W}{g}\frac{\dot{x}^2}{2} = 2.294\,\dot{x}^2$

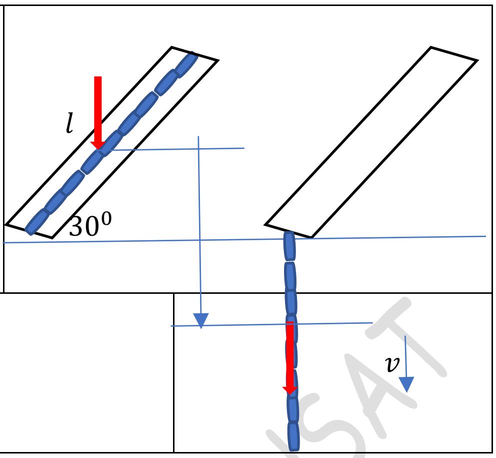


Change in kinetic energy $= 2.294\,\dot{x}^2 - 20.64$
Net work done by the force over the displacement $= (W \sin 30 - \mu R) \times 15$
$$= (W \sin 30 - \mu W \cos 30) \times 15 = 162.13\ Nm$$
By work-energy principle, change in kinetic energy=net work done
$$2.294\,\dot{x}^2 - 20.64 = 162.13$$
$$\dot{x} = 8.93\ m/s$$

Solve this problem by Newton's law/DÁlembert's principle.

Example 3

A length l of a smooth straight pipe held with its axis inclined to the horizontal by an angle of 30^0 contains a flexible chain also of length l and weight W. Neglecting friction and assuming that, after release the chain falls vertically as it emerges from the open end of the pipe, find the velocity v with which it leaves the pipe.



The initial kinetic energy of the chain $= 0$

Final kinetic energy of the chain $= \frac{W}{g}\frac{v^2}{2}$

Change in kinetic energy $= \frac{W}{g}\frac{v^2}{2} - 0 = \frac{W}{g}\frac{v^2}{2}$

Since the friction is neglected, the only force acting is the weight W.

The centre of mass of the chain has a displacement

$$= \frac{l}{2}\sin 30 + \frac{l}{2} = \frac{3l}{4}$$

The work done by the weight $= W.\frac{3l}{4}$

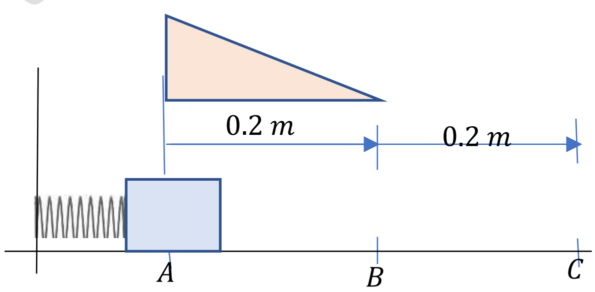
By the work-energy principle, change in kinetic energy is equal to the work done.

$$\frac{W}{g}\frac{v^2}{2} = W.\frac{3l}{4}$$

$$v = \sqrt{\frac{3}{2}gl}$$

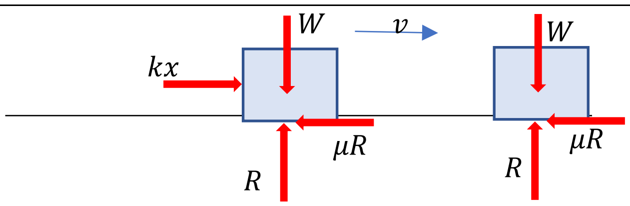
Example 4

A $98\ N$ block rests on the horizontal surface as shown. The spring which is not attached to the block has a stiffness $k = 500\ N/m$ and is initially compressed $0.2\ m$ from B to A. After the block is released from rest from A, determine its velocity when it passes point C. The coefficient of kinetic friction between the block and the plane is $\mu = 0.2$



Since the spring is not attached to the block, the spring force will be acting on the block only up to B. The free-body diagrams of the block when it is before and after point B.

The initial kinetic energy of the block at A =0



Let the velocity of the block as it reaches point C be v.

The final kinetic energy of the body $= \frac{W}{g}\frac{v^2}{2}$

In the y-direction, $\Sigma F_y = 0 \ggg \ R = W$

Work done by spring force $= ½\, kx^2 = ½ \times 500 \times 0.2^2 = 10\ Nm$

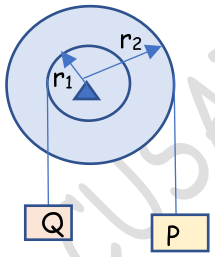
Work done by frictional force$= -\mu R \times 0.4 = -0.2 \times 98 \times 0.4 = -7.84\ Nm$

Net work done by the forces$= 10\ Nm - 7.84\ Nm = 2.16\ Nm$

By work energy principle, change in kinetic energy= net work done

$$\frac{W}{g}\frac{v^2}{2} = 2.16$$

$$v = 0.656\ m/s$$

#Exercise 1 A gun weighing $668\ kN$ fires a $4.5\ kN$ shell with a muzzle velocity $1100\ m/s$. The gun is nested in springs having a total spring constant $k = 27\ MN/m$. Assuming that the explosion is over before the gun has a chance to move perceptibly, how far will it recoil after the explosion?	
#Exercise 2 If the system in figure is released from rest in the configuration shown, find the velocity of the falling weight P when it has travelled a distance of $10\ m$. Given $P = Q = 50\ N$, $r_1 = 10\ cm, r_2 = 15\ cm$	
#Exercise 3 A package is projected 10 m up a 15^0 incline so that it just reaches the top of the incline with zero velocity. Knowing that the coefficient of kinetic friction between the package and the incline is 0.12, determine (a) the initial velocity of the package at A, (b) the velocity of the package as it returns to its original position.